

SATYA BHARGAV MADDIPATI

📍 Madison, WI 📞 +1-608-298-8982 ✉ sbmaddipati@wisc.edu [LinkedIn](#) [Portfolio](#) [GitHub](#)

Education

University of Wisconsin-Madison

M.S. in Computer Sciences | GPA: 4.0/4.0

Madison, WI

Sep 2025 - May 2027 (Expected)

Indian Institute of Technology Bombay

B.Tech-M.Tech Dual Degree with Honors | GPA: 8.9/10

Mumbai, India

Aug 2018 - Jun 2023

Experience

Tata AIA Life Insurance | Assistant Manager, Data Science (prev. Mgmt. Trainee)

Mumbai, India

• [PYTHON](#) • [FASTAPI](#) • [MICROSERVICES](#) • [DOCKER](#) • [AZURE](#) • [CI/CD](#)

Jul 2023 - Dec 2025

- Built a **GPT-4/Azure OpenAI RAG support system** with **FastAPI** microservices, serving **50K+** users at **0.8 s p95 latency** and **99.5% uptime**; deployed on **Azure Container Apps**.
- Designed and integrated **LangGraph/CrewAI workflows** into insurance claims processing, reducing manual review effort by **35%** and saving an estimated **2,000+ person-hours annually**.
- Fine-tuned **5+ open-source LLMs (Llama 3.2, Qwen 2.5)** with **QLoRA/PEFT**, improving document-classification accuracy from **78% to 94%**; made training reusable and configurable with automated tests, CI/CD, and peer review.
- Designed a modular insurance recommendation service over **2M+ customer records** using **embeddings and collaborative filtering**; exposed predictions through **REST APIs consumed by internal teams**.

IIT Bombay | Graduate Research Assistant

Mumbai, India

Time-Series Modeling, Forecasting, and Interpretability

Apr 2022 - Jun 2023

- Modeled **multiyear air-quality and meteorological time series** from Sion CAAQMS in Mumbai; applied **KZ filtering and multiple linear regression** to separate weather-driven variation from underlying pollutant trends.
- Benchmarked **SARIMA and grid-search-tuned Random Forests** across raw/detrended signals and forecast horizons; used **SHAP** to interpret **humidity, wind, and rainfall** contributions to predictions.

Eureka Infotech Inc

Mumbai, India

Data Analyst Intern — AWS Batch, Docker, AutoML

May 2021 - Jun 2021

- Re-engineered **AWS Batch** forecasting workflows, reducing runtime from **2.5 days to 145 minutes**; **Dockerized an AutoML framework** integrating multiple models and hyperparameter optimization.

Projects

Buzzer: Interactive Quiz Platform | [REACT](#) • [TYPESCRIPT](#) • [FASTAPI](#) • [SOCKET.IO](#)

Sep 2026 - Present

- Extended a multi-user **React/TypeScript, FastAPI, and Socket.IO** quiz platform across **Admin, Host, and Player apps**, implementing two question types with shared partial-credit scoring, aspect-ratio-correct canvas interactions, and REST/socket payload contracts.
- Added request/answer validation and integration tests spanning **API payloads, answer persistence, import/export, and session reports**; replaced silent question-type fallbacks in authoring and scoring with explicit errors.

Reason-Guessr: Interpretable Image Geolocation | [CLIP ViT-B/32](#)

Live application

- Designed an interpretable geolocation model combining **CLIP ViT-B/32 embeddings** with **climate, license-plate, vegetation, and architecture cues** for continuous coordinate prediction.
- Evaluated on **32K images across 8,000 locations**, reducing median haversine error from **90.6 km to 88.9 km**; deployed an interactive web application with a browser frontend and **Python inference service**.

Technical Skills

Programming: C++, TypeScript/JavaScript, Python, Java, SQL

Software: FastAPI, React, REST APIs, Git, unit/integration testing, PostgreSQL, MySQL, OOP

AI/ML: PyTorch, Hugging Face, Scikit-learn, CLIP, XGBoost, SHAP, LangGraph, CrewAI, QLoRA/PEFT

Cloud: Azure Container Apps, AWS Batch, Docker, CI/CD

Achievements

Academic: JEE Advanced Top 0.5% among 250K candidates.

Activities: **Finalist**, Tata Innovation Workshop (**25+** teams) — WhatsApp AI cross-sell flow cutting add-on processing from **15 days to under 5 min**; Web Coordinator, Techfest IIT Bombay; Sports Councilor, IIT Bombay; Inter-IIT Sports Bronze Medal.